The Testbed Network

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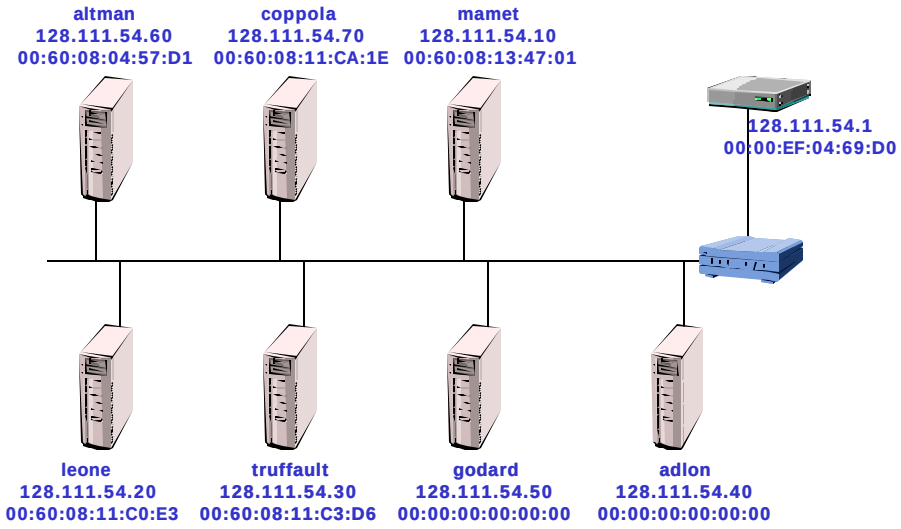
The Testbed

Structure

- Testbed IP address are allocated in subnet 128.111.54
- Flat topology: hosts are connected by Ethernet hub
- Default gateway 128.111.54.1
- Currently available hosts have Linux Redhat 6.2 installed
- Additional hosts with Solaris and Windows NT will be available soon

Topology



Access to the Testbed

- Testbed hosts can be accessed from *anywhere* through Secure Shell (ssh)
- Traffic is allowed only to/from the following networks:
  - 128.111.54 (testbed)
  - 128.111.49 (GSL)
  - 128.111.43 (CISL)
  - 128.111.51 (ECI)
- Initially available hosts: coppola, altman, mamet, leone
- As soon as you log in, CHANGE YOUR PASSWORD!
  - passwd *yourhandle*
- Truffault is a special host: DO NOT TOUCH!!

Access to the Testbed

- Before starting a session in the testbed from host with IP address REMOTE_IP send an email
  - *From: your CS account*
  - To: vigna@cs.ucsb.edu
  - Subject: TESTBED BEGIN YYMMDDHHMM *HANDLE* REMOTE_IP
  - *No body* (meaning: no signature, no data)
- Ssh into a host in the testbed using your handle
  - ssh masterzero@coppola.cs.ucsb.edu
- When inside the testbed you can use telnet, rsh, etc to move inside the testbed

Access to the Testbed

- DO NOT USE TELNET FROM/TO THE OUTSIDE
- DO NOT USE FTP FROM/TO THE OUTSIDE
- Read and understand the SSH man page
- To transfer files from/to the outside use *always* Secure Copy (scp)
  - From the outside to the testbed
    % scp file handle@testbedhost:
  - From the inside to the outside
    % scp handle@testbedhost:file .
- Your CS password must never be used while on the testbed

Access to the Testbed

- When you are finished send an email
  - *From: your CS account*
  - To: vigna@cs.ucsb.edu
  - Subject: TESTBED END YYMMDDHHMM *HANDLE* REMOTE_IP
  - *No body* (meaning: no signature, no data)

Example

- Mark Twain (CS account: *mtwain,* handle: *bzero*) starts a session at 8:32am Jan 31 from host 128.111.48.69 (longboard.cs.ucsb.edu)
- Email
  - From: mtwain@cs.ucsb.edu
  - To: vigna@cs.ucsb.edu
  - Subject: TESTBED BEGIN 0101310832 bzero 128.111.48.69
- Login to coppola

  longboard% ssh bzero@coppola

  bzero@coppola's password:

  [bzero's password (not your CS password!)]

  Last login: Sun Jan 28 19:35:35 2001

  [...]

Example

- Moving file exploit.c to coppola

  longboard% scp exploit.c bzero@coppola:

  bzero@coppola's password:

  exploit.c | 1 KB | 0.0 kB/s | ETA: 00:00:00 | 100%
- Compiling the file

  coppola> gcc exploit.c -o exploit
- Running the program

  coppola> ./exploit > output.txt
- Moving output.txt back to longboard

  longboard% scp bzero@coppola:output.txt .

  bzero@coppola's password:

  output.txt | 2 KB | 0.0 kB/s | ETA: 00:00:00 | 100%

Example

- Mark Twain closes the session

  coppola> logout
- Email (it's 2:17pm = 14:17)
  - From: mtwain@cs.ucsb.edu
  - To: vigna@cs.ucsb.edu
  - Subject: TESTBED END 0101311417 bzero 128.111.48.69

I WANT TO BE ROOT!

Acceptable Use

- The network is to be used for security testing only
  - It's not a storage space
  - It's not a development environment
- Resources are limited and there are many users
  - Try to limit your resource usage (disk, network, CPU)
- Be sure you know what you are doing
  - Do not execute programs whose behavior is unknown
  - Do not flood the network
    - Do not perform denial-of-service attacks for extended amounts of time
    - Do not start an attack if you don't know how to stop it!
  - Do not remove/access other users' files or change host configuration files

Acceptable Use

- Do not use the testbed to attack outside hosts
- Do not store anything important on the testbed
  - The hosts may go down at any moment
  - The network may go down at any moment
  - The disk may be formatted at any moment
- Please, try to leave the network as you found it
  - No trojan horses
  - No permanent side effects
  - No software installed in commonly used path (/bin, /usr/bin, /usr/local/bin, etc)
- Try to use your brains and have fun!

What If Something Goes Wrong?

- First try to understand what went wrong
  - Is it something related to your activities?
- Send email from your CS account
  - to vigna@cs
  - CC to chemdog3@cs
  - CC to ellisd@cs
- Try to explain what happened
- Try to be clear and concise
  - YES: “host coppola does not respond to ping from host 128.111.49.12 [trace follows]”
  - NO: “The network doesn’t work”

Caveats

- Expect a lot of frustration
- This is an experiment!
- You are many
- Teacher and TAs have a life of their own
- Be patient
- Check the class web page for updates